

Chemical Bonding: Hybridisation and Spatial Geometry:

Source questions

Q1 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Electron-Domain Counting and Molecular Shape Geometry

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric Number and Hybridisation in Molecules with Lone Pairs

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

BF₃ and NH₃ each have three sigma bonds to the central atom. BF₃ has no central lone pair; NH₃ has one. Explain why three

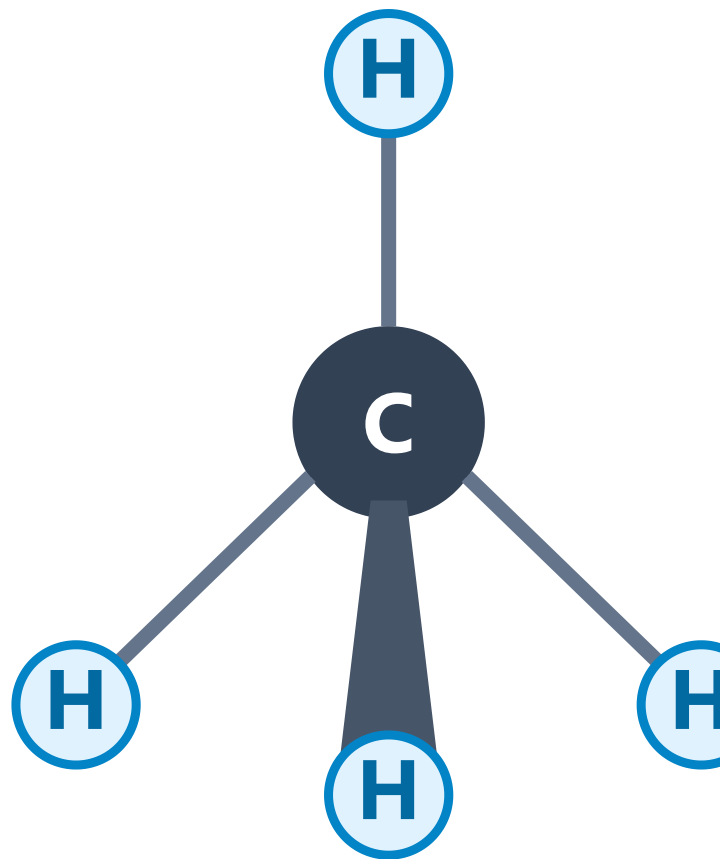
Representation

sigma bonds alone cannot determine the same central hybridisation and molecular shape for both.

Common wrong route · open if you want a warning

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CH₄ (0 Lone Pairs)



Tetrahedral

Bond Angle: 109.5°
Domain geom = Shape

CH₄

Need a hint? · guided support

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Need the concept again?

Electron-Domain Counting and Molecular Shape Geometry

Q2 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Multiple Bonds, Orthogonal Pi Systems, and Terminal Atom Hybridisation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For $\text{H}_2\text{C}=\text{O}$, assign the carbon hybridisation, count the sigma and pi components of the $\text{C}=\text{O}$ bond, and identify the carbon orbital left available for the pi component. Keep electron-domain counting separate from counting shared electron pairs.

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Need the concept again?

Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Common wrong route · open if you want a warning

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Q3 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Multiple Bonds, Orthogonal Pi Systems, and Terminal Atom Hybridisation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

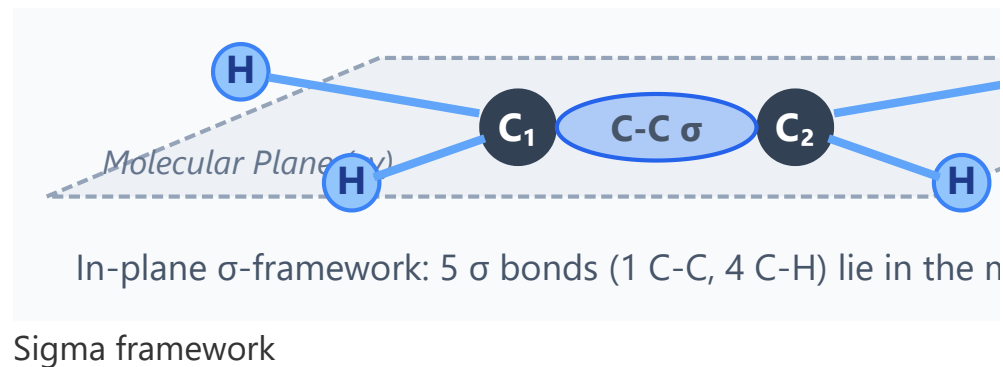
ATTEMPT FIRST

Ethene is $\text{H}_2\text{C}=\text{CH}_2$. Translate this Lewis representation into a qualitative orbital-overlap diagram showing the C–C sigma framework, the two unhybridised carbon p orbitals and their side-on overlap. Label which directions are in the molecular plane and which are perpendicular to it.

Common wrong route · open if you want a warning

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Representation



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Need the concept again?

Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Q4 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Multiple Bonds, Orthogonal Pi Systems, and Terminal Atom Hybridisation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For CO₂, connect the central-carbon Lewis domain count to its hybrid-orbital model, the number of remaining carbon p orbitals and the sigma/pi composition of its two C=O bonds. Explain how these pieces fit one consistent linear arrangement.

Representation

Common wrong route · open if you want a warning

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Stage 1: Lewis Domain Cour
:Ö = C = Ö: → 2 Electron Do
Double bond = 1 domain; Linea

Domains

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Need the concept again?

Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Q5 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Electron-Domain Counting and Molecular Shape Geometry

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric Number and Hybridisation in Molecules with Lone Pairs

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Consider the claim: 'Any central atom with three sigma bonds must be sp^2 .' Given BF_3 : three B–F sigma bonds, zero boron lone pairs, planar; NH_3 : three N–H sigma bonds, one nitrogen lone pair, pyramidal. Decide whether these observations support the claim, give the decisive warrant, and state a more defensible introductory rule. Explain what the shape observations alone do not prove about orbitals.

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Need the concept again?

Electron-Domain Counting and Molecular Shape Geometry

Common wrong route · open if you want a warning

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Q6 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Electron-Domain Counting and Molecular Shape Geometry

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric Number and Hybridisation in Molecules with Lone Pairs

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

In CH₃F, carbon has three C–H single bonds and one C–F single bond. Assign its introductory local hybridisation and explain why the number of electron-domain directions is unchanged from CH₄ even though the attached atoms are different. Do not assume all bond properties are identical.

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Need the concept again?

Electron-Domain Counting and Molecular Shape Geometry

Common wrong route · open if you want a warning

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Q7 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Multiple Bonds, Orthogonal Pi Systems, and Terminal Atom Hybridisation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

HCN is represented as $\text{H}-\text{C}\equiv\text{N}$ with a lone pair on nitrogen. Assign the introductory hybridisation at carbon and nitrogen, and count the sigma and pi components of the

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Need the concept again?

Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

$\text{C}\equiv\text{N}$ bond. Explain how the local domain count is compatible with those components.

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Q8 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

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Concept: Electron-Domain Counting and Molecular Shape Geometry

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric Number and Hybridisation in Molecules with Lone Pairs

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

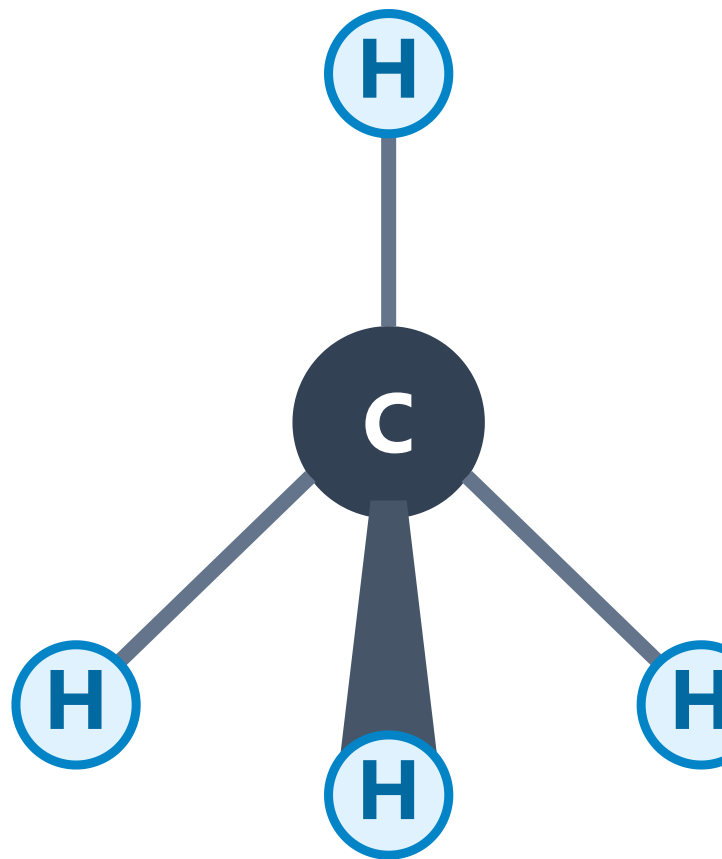
Representation

For methylamine, $\text{CH}_3\text{-NH}_2$, carbon has four single bonds and nitrogen has three single bonds plus a lone pair. Compare the introductory local hybridisation and arrangement of bonded atoms at carbon and nitrogen. Explain why the same hybridisation label does not require identical molecular shapes.

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CH_4 (0 Lone Pairs)



Tetrahedral

Bond Angle: 109.5°
Domain geom = Shape

CH₄

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Need the concept again?

Electron-Domain Counting and Molecular Shape Geometry

Q9 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Multiple Bonds, Orthogonal Pi Systems, and Terminal Atom Hybridisation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A carbon centre has three sigma-bond directions, no lone pair, and participates in one pi bond. Translate this description into a qualitative local orbital arrangement and check that the orbitals used in the sigma and pi parts have not been counted twice.

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Need the concept again?

Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Common wrong route · open if you want a warning

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Q10 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Electron-Domain Counting and Molecular Shape Geometry

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric Number and Hybridisation in Molecules with Lone Pairs

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A student says, 'If the electron-domain geometry is tetrahedral, the atoms must form a tetrahedral molecular shape.'

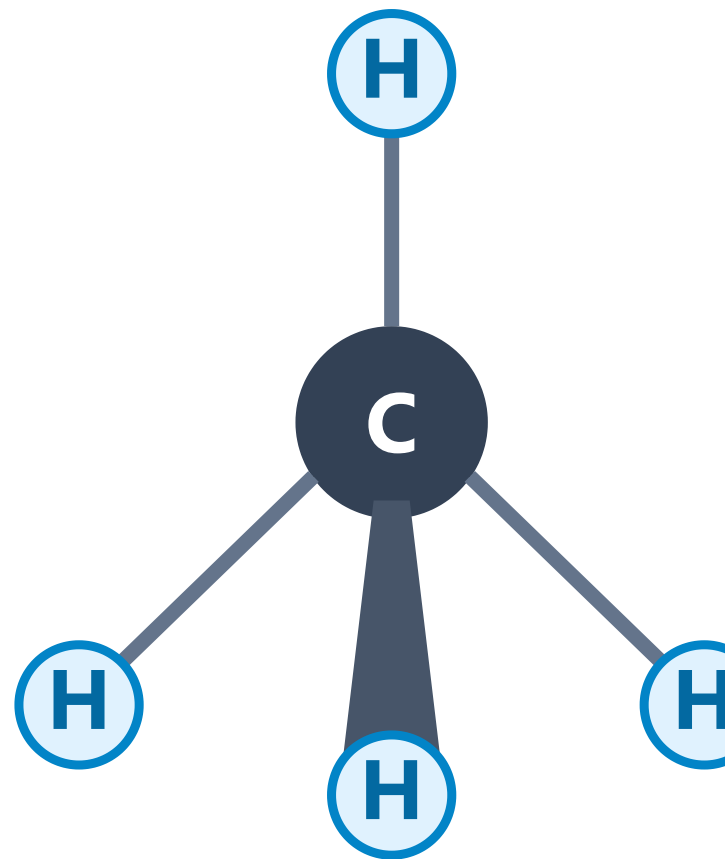
Evaluate the statement using CH_4 , NH_3 and H_2O , with their respective central lone-pair counts of zero, one and two.

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Representation

CH₄ (0 Lone Pairs)



CH₄

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Need the concept again?

Electron-Domain Counting and Molecular Shape Geometry